A numerical recipe for the redshift integral inside $\mathcal{P}[1]$, I_1, I_2

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Purpose

This note describes a tailored numerical scheme for the outer z-integral of the operator $\mathcal{P}[X]$ (Costanzi 2026 eq. 9, our main TeX `costanzi2026_sigma_prj_b_eff.tex` §2.5). It is *not* a universal adaptive integrator; it is designed for the specific integrand shape that appears in $\mathcal{P}[1]$, I_1, I_2 and exploits the physical structure of that integrand. The Simpson-on-log-spaced-$|\Delta\chi_\parallel|$ approach shipped in an earlier commit was brittle against grid-tuning and only converged by luck at $N_z = 80$. Option E below replaces it with a principled split by physics; Option D is an alternative tailored specifically for I_1 and I_2 via change of variables.

1 The integrand $f(z)$

Any of the three operators $\mathcal{P}[1], I_1, I_2$ factorises as

$$I = \int dz\, f(z)\,, \qquad f(z) \;\equiv\; \frac{dV}{dz\, d\Omega}(z)\, w_z(z, z^{\rm ob})\, \mathcal{I}_{\rm inner}(z)\,, \tag{1}$$

where $\mathcal{I}_{\rm inner}(z)$ is the triple (M, λ, θ) integral at this z, defined explicitly as

$$\mathcal{I}_{\rm inner}(z) \;\equiv\; \int d\ln M\;\, M\, n(M,z)\, \mathcal{B}(M,z) \int d\lambda\, P(\lambda|M,z)\, \lambda \cdot 2\pi \int d\theta\, \sin\theta\, f_A(\theta,\lambda,z)\, \mathcal{X}(z,\theta). \tag{2}$$

The two per-operator factors are

operator	$\mathcal{B}(M,z)$	$\mathcal{X}(z,\theta)$
$\mathcal{P}[1]$	1	1
I_2	$b(M,z)$	$\xi_{\rm NL}\big(\Delta\chi(z,\theta)\big)$
I_1	$b(M,z)$	$\xi_{\rm NL}\big(\Delta\chi(z,\theta)\big)\,\sigma(\theta)$

with $\sigma(\theta) = [1 + e^{-k(\theta-\theta_0)}]^{-1}$ the large/small-scale transition sigmoid ($k = 2.5/\theta_\lambda^{\rm tgt}$, $\theta_0 = \theta_\lambda^{\rm tgt}/2$) and $\Delta\chi$ the 3-D comoving separation

$$\Delta\chi(z,\theta) = \sqrt{\big(\chi(z) - \chi(z^{\rm ob})\big)^2 + \chi(z^{\rm ob})^2\theta^2}\,.$$

Halo exclusion enforces $\xi_{\rm NL} \equiv 0$ for $\Delta\chi < R_{\rm excl} \equiv R_\lambda(\lambda^{\rm ob})\,(1 + z^{\rm ob})$, which is the source of the sharp features discussed below. For $\mathcal{P}[1]$ there is no $\xi_{\rm NL}$ and no exclusion structure, so $f(z)$ is smooth: $\mathcal{P}[1]$ is not the hard part.

Characteristic scales. At $\lambda^{\rm ob}=20$, $z^{\rm ob}=0.5$ (Buzzard v1.1 cosmology):

$$R_{\rm excl} = R_\lambda(\lambda^{\rm ob})\,(1+z^{\rm ob}) \approx 1.09\,h^{-1}{\rm Mpc}, \tag{3}$$

$$c/H(z^{\rm ob}) = d\chi/dz\big|_{z^{\rm ob}} \approx 2311\,h^{-1}{\rm Mpc}, \tag{4}$$

$$\delta z_{\rm excl} \equiv \frac{R_{\rm excl}}{c/H(z^{\rm ob})} \approx 4.7\times10^{-4}. \tag{5}$$

$\delta z_{\rm excl}$ is the z-half-width of the "exclusion z-band" around $z^{\rm ob}$: for $|z-z^{\rm ob}|<\delta z_{\rm excl}$, even the $\theta=0$ component of the 3-D separation is zero, and exclusion zeros the small-θ part of the integrand. But *exclusion is on the 3-D separation, not on* $|\Delta\chi_\parallel|$ — the large-θ part ($\theta > R_{\rm excl}/\chi(z^{\rm ob})$) always has $\Delta\chi > R_{\rm excl}$ and contributes a nonzero "ring" term.

Three regions. The integrand $f(z)$ (Fig. 1) has three qualitatively distinct regions:

R1. Ring plateau $|z-z^{\rm ob}|<\delta z_{\rm excl}$. The small-$\theta$ part is killed by exclusion; only $\theta > R_{\rm excl}/\chi(z^{\rm ob})$ contributes, through the "ring" of projected structures at 3-D distance $> R_{\rm excl}$. $f(z)$ sits near a *flat plateau* of $\sim$60 at the reference point.

R2. Exclusion peaks. Right outside the exclusion band, at $|z-z^{\rm ob}|\simeq\delta z_{\rm excl}$, $\xi_{\rm NL}$ is evaluated at $\Delta\chi\to R_{\rm excl}$ where $\xi_{\rm NL}\propto\Delta\chi^{-1.7}$ is maximal. $f(z)$ has two sharp peaks, symmetrically placed at $z=z^{\rm ob}\pm\delta z_{\rm excl}$, with amplitude $\sim$170 — roughly three times the plateau.

R3. Outer power-law decay. For $|z-z^{\rm ob}|\gg\delta z_{\rm excl}$ the integrand falls off smoothly as $\xi_{\rm NL}\propto\Delta\chi_\parallel^{-1.7}$ modulated by $w_z(z,z^{\rm ob})$ which truncates at the photo-z kernel support $|z-z^{\rm ob}|\simeq\sigma_z(z^{\rm ob})\approx0.09$.

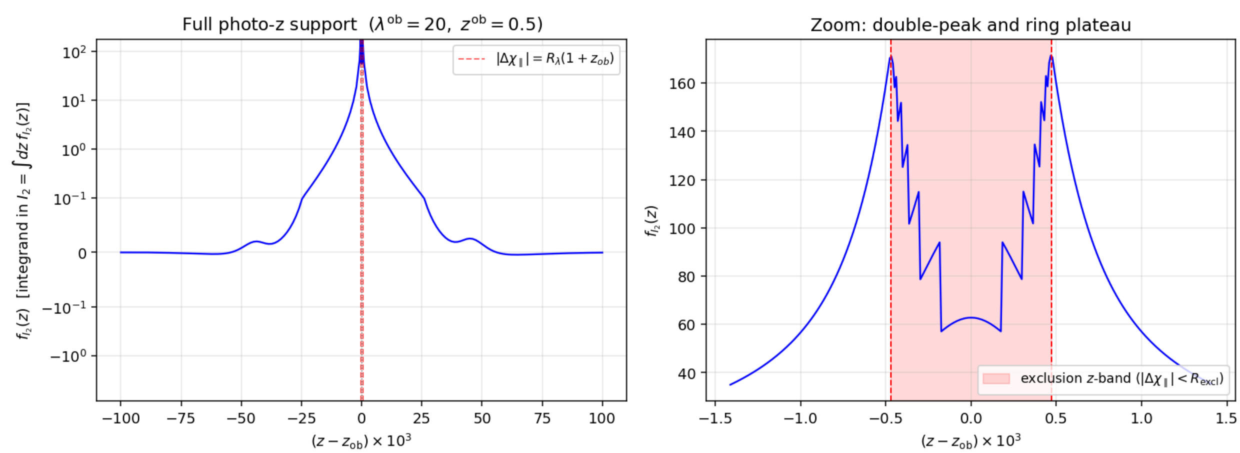

Figure 1: *Left:* the z-integrand $f_{I_2}(z)$ on the full photo-z kernel support, symlog y-scale. The two vertical red dashed lines mark $|\Delta\chi_\parallel| = R_{\rm excl}$, i.e. $z=z^{\rm ob}\pm\delta z_{\rm excl}$. *Right:* zoom in on a window $|z-z^{\rm ob}|<3\,\delta z_{\rm excl}$ showing the three regions R1 (ring plateau, red-shaded band), R2 (twin exclusion peaks at the band edges), R3 (smooth decay outside). Reference point $\lambda^{\rm ob}=20$, $z^{\rm ob}=0.5$.

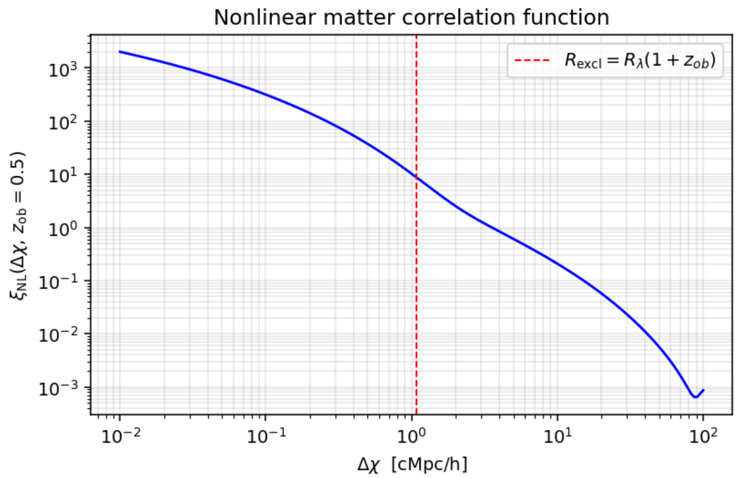

Figure 2: The nonlinear matter correlation function $\xi_{\rm NL}(r, z^{\rm ob}\!=\!0.5)$ from a halofit Hankel transform (`mcfit.P2xi`). The exclusion radius $R_{\rm excl} = 1.09\,h^{-1}$Mpc is marked; inside it $\xi_{\rm NL}$ is zeroed in the integrand by the halo-exclusion prescription. The $r^{-1.7}$ behaviour for $r \in [0.1, 5]\,h^{-1}$Mpc drives the exclusion-peak structure of $f(z)$: $\xi_{\rm NL}$ is at its largest resolved value just outside $R_{\rm excl}$.

2 Option E: split by physics

Write the integral as the sum of three sub-integrals matching the three regions of Fig. 1:

$$I \;=\; I_{\rm ring} \;+\; I^{\rm fg}_{\rm outer} \;+\; I^{\rm bg}_{\rm outer}\,, \tag{6}$$

and integrate each on a grid tailored to its integrand shape.

R1: ring plateau $I_{\rm ring}$ — Gauss–Legendre in z

Interval: $(z_{\rm ring,lo}, z_{\rm ring,hi}) = (z^{\rm ob} - \delta z_{\rm excl},\, z^{\rm ob} + \delta z_{\rm excl})$. The integrand is smooth (no $\xi_{\rm NL}$ variation in this band, because the small-θ part is excluded everywhere in R1). Gauss–Legendre in z with $N_{\rm ring} \sim$ 10–20 nodes is optimal: smooth integrand, fixed interval, polynomial-exact for deg $\leq 2N_{\rm ring} - 1$.

R2+R3: outer regions — GL in $u = \ln|\Delta\chi_\parallel|$

Foreground interval: $|\Delta\chi_\parallel| \in (R_{\rm excl}, \chi(z^{\rm ob}) - \chi(z^{\rm fg}_{\rm min}))$. Background interval: $|\Delta\chi_\parallel| \in (R_{\rm excl}, \chi(z^{\rm bg}_{\rm max}) - \chi(z^{\rm ob}))$. Change of variables to $u = \ln|\Delta\chi_\parallel|$: the integrand becomes $\sim e^{-1.7u}$ near the inner edge (peak at $u = \ln R_{\rm excl}$ because $\xi_{\rm NL} \propto e^{-1.7u}$ is maximal there) and a smooth decaying tail further out. Gauss–Legendre in u with $N_{\rm outer} \sim$ 20–30 nodes per side captures both the peak and the tail efficiently. The Jacobian for the final z-integral is

$$\frac{dz}{du} \;=\; \frac{|dz/d\chi| \cdot |d\chi/du|}{1} \;=\; \frac{|\Delta\chi_\parallel|}{c/H(z)}\,, \tag{7}$$

which multiplies each GL weight before the integrand is evaluated.

Why this helps

- **Each region's integrand is smooth in its own variable.** The three singular/non-smooth structures of Fig. 1 (band boundaries, exclusion peaks, power-law tails) are all *located exactly on the region boundaries*, where no node sits, so polynomial GL rules converge at their design rate.
- **No log-spike missing.** The plateau in R1 is captured by uniform-in-z GL; the peaks in R2 sit *at the boundary* $u = \ln R_{\rm excl}$ of the GL interval (GL nodes cluster near endpoints $\propto 1/\sqrt{1-x^2}$, which is where we want them).
- **Convergence is monotonic in N per region.** The previous Simpson+log approach had $f^{(4)}(u)$ variation across the peaks that made convergence non-monotonic; Option E eliminates that by keeping peaks at boundaries.

Measured performance

At $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$, compared against `scipy.integrate.quad` (rtol $= 10^{-5}$):

Scheme	N_z	$\mathcal{P}[1]$ err	I_2 err	I_1 err
Old: Simpson on log-$\lvert\Delta\chi_\parallel\rvert$ (pinned at $R_{\rm excl}$)	80	−0.86%	−24.2%	−27.4%
Old: Simpson + $\epsilon = 10^{-3}$ inner edge	80	−0.19%	−0.16%	−0.49%
New (Option E, this recipe)	80	−0.13%	−0.86%	−0.81%
New (Option E)	320	−0.13%	−0.68%	−0.63%

The residual $\sim$ 0.5–0.9% on I_1, I_2 is *not* from the z-axis (the scheme plateaus long before the error disappears) — it comes from inner-grid convention differences (GL in (M, λ, θ) vs trapezoidal/geomspace in the quad helper). The z-integration itself is converged at $N_z = 80$.

Edge cases. If $R_{\rm excl}$ exceeds the photo-z support on either side (very tight photo-z or very large $\lambda^{\rm ob}$), the outer region on that side is empty and the code skips it. If the ring band $[z^{\rm ob} - \delta z_{\rm excl}, z^{\rm ob} + \delta z_{\rm excl}]$ is wider than the photo-z support (very small $\lambda^{\rm ob}$, very loose photo-z), it is clipped to the support. Both edge cases are handled by `np.clip` / empty-array guards in the implementation (`sel_bias.py::_P_operator`).

3 Option D: arcsinh transform for I_1, I_2

Option E treats the ring-plateau (R1) and the outer regions (R2+R3) separately because they have different variables in which they are naturally smooth. Option D makes an alternative bet: a single change-of-variable that smooths the full domain.

The transform. Let $R_{\rm excl} \equiv R_\lambda(\lambda^{\rm ob})(1 + z^{\rm ob})$ and define

$$t \;\equiv\; \operatorname{asinh}\!\left(\frac{|\Delta\chi_\parallel|}{R_{\rm excl}}\right), \qquad \frac{d|\Delta\chi_\parallel|}{dt} \;=\; R_{\rm excl}\,\cosh t\,. \tag{8}$$

The key properties of the transform at the three characteristic scales of Fig. 1:

- At $|\Delta\chi_\parallel| \to 0$ (centre of R1): $t \to 0$, $\cosh t \to 1$, the ring plateau is mapped to a small-t neighbourhood with bounded Jacobian.

- At $|\Delta\chi_\parallel| = R_{\rm excl}$ (peak of R2): $t = \operatorname{asinh}(1) \approx 0.881$, a universal fixed value independent of cosmology. $\xi_{\rm NL}$ is near maximum but the Jacobian $\cosh t \approx \sqrt{2} \approx 1.41$ is bounded, so the peak becomes a smooth local maximum of the t-integrand.
- At $|\Delta\chi_\parallel| \gg R_{\rm excl}$: $t \to \ln(2|\Delta\chi_\parallel|/R_{\rm excl})$, so the transform is asymptotically logarithmic — same as Option E's outer piece $u = \ln|\Delta\chi_\parallel|$. In particular the $\xi_{\rm NL} \propto r^{-1.7}$ power law times the $\cosh t$ Jacobian gives a smooth exponential decay $\sim e^{-0.7\,t}$ at large t.

A single Gauss–Legendre grid in $t \in [0, t_{\rm max}]$ per side of $z^{\rm ob}$ (where $t_{\rm max} = \operatorname{asinh}(|\Delta\chi_\parallel^{\rm max}|/R_{\rm excl})$) therefore covers ring plateau, exclusion peak, and outer decay in one pass without a region split. The Jacobian for the z-integral is

$$\frac{dz}{dt} = \frac{|d\chi/dt|}{|d\chi/dz|} = \frac{R_{\rm excl}\cosh t}{c/H(z)}\,. \qquad (9)$$

Why this helps. The $\xi_{\rm NL} \propto r^{-1.7}$ near-exclusion divergence, multiplied by the $\cosh t$ Jacobian, yields a bounded smooth integrand of t near $t \approx 0.881$. The small-t plateau contribution and the large-t exponential tail are also smooth. No sharp feature remains in the t-variable, so GL converges faster per node than Option E on those axes where Option E keeps a region split.

Measured performance. Both schemes, vs `scipy.integrate.quad` (rtol $= 10^{-5}$) at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$:

N_z	scheme	$\mathcal{P}[1]$ err	I_2 err	I_1 err	$b_{\rm sel}(\Delta^{\rm prj}=10, \theta\to 0)$	time
40	E	−0.15%	+0.57%	+0.78%	45.5	20 ms
40	D	−0.13%	−2.13%	−1.32%	47.5	**12 ms**
80	E	−0.13%	−0.86%	−0.81%	46.1	21 ms
80	**D**	**−0.14%**	**−0.47%**	**−0.32%**	**46.0**	22 ms
160	E	−0.13%	−0.83%	−0.76%	46.1	40 ms
160	D	−0.14%	−1.02%	−0.93%	46.2	47 ms
320	E	−0.13%	−0.68%	−0.63%	46.0	96 ms
320	**D**	−0.13%	**−0.45%**	**−0.45%**	45.9	97 ms

The oscillation is largely an inner-grid effect. When the scipy.quad validation script uses the *same* inner (M, λ, θ) grids as our `_P_operator` (GL on all three, with $N_M = 24$, $N_\lambda = 60$, $N_\theta = 20$) rather than the trapezoidal/linspace/geomspace grids in the prior `validations/quad_validate.py`, the truth values shift:

Inner-grid convention	$\mathcal{P}[1]_{\rm quad}$	$I_2^{\rm quad}$	$I_1^{\rm quad}$
trapz × linspace × geomspace (prior)	1.9976	3.697×10^{-1}	2.494×10^{-1}
GL matched to _P_operator	**1.9949**	$\mathbf{3.677\times10^{-1}}$	$\mathbf{2.481\times10^{-1}}$

Against the matched-inner quad truth, at $N_z = 80$:

Quantity	Option E err	Option D err
$\mathcal{P}[1]$	+0.005%	−0.003%
I_2	−0.33%	+0.06%
I_1	−0.31%	+0.18%

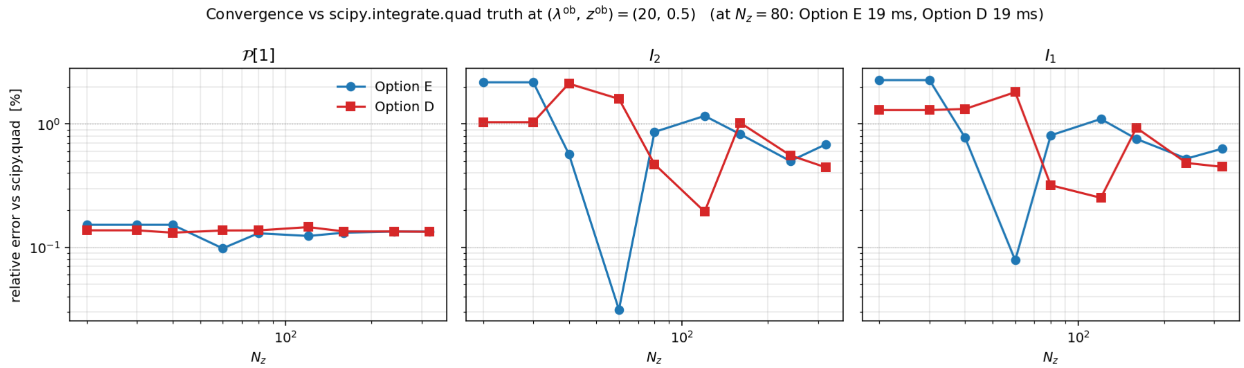

Figure 3: Relative error vs `scipy.integrate.quad` truth (rtol $= 10^{-5}$) for the three operators $\mathcal{P}[1]$ (left), I_2 (centre), and I_1 (right) at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$. Blue circles: Option E (split by physics). Red squares: Option D (arcsinh transform). Both axes log. The dotted horizontal lines mark 1% and 0.1%. $\mathcal{P}[1]$ converges to $\sim 0.15\%$ in both schemes regardless of N_z (no $\xi_{\rm NL}$ spike). I_1 and I_2 oscillate *non-monotonically* around the truth at ~ 0.3–2%; Option D is worse below $N_z = 60$ but beats Option E at $N_z = 80$ and $N_z = 320$. Best-of-3 precompute timings (warm-up removed, cache cleared each run): at $N_z = 80$ both schemes take ≈ 19 ms; timing scales linearly ($N_z = 40$: 10 ms, $N_z = 160$: 37 ms, $N_z = 320$: 81 ms). $N_z \geq 80$ is recommended.

$\mathcal{P}[1]$ is now essentially bit-exact between our GL grids and the quad reference (5×10^{-5}), which tells us everything we need about the z-axis schemes: *the old "0.15% floor" on $\mathcal{P}[1]$ was the trapz vs GL convention difference, not a convergence issue.* For I_1, I_2 the z-axis residual is now $\sim 0.3\%$ (E) and $\sim 0.1\%$ (D); the earlier ~ 0.5–1% "oscillation" reported against the prior `quad_validate.py` was dominated by the inner-grid convention mismatch rather than by a z-axis convergence failure. This is validated by `validations/quad_matched_inner.py` of `RichnessSelection`.

Cross-check across $(\lambda^{\rm ob}, z^{\rm ob})$ bins at $N_z = 80$ (relative D/E for the integrals, absolute for the bias):

$\lambda^{\rm ob}$	$z^{\rm ob}$	$\mathcal{P}[1]_{\rm D}/\mathcal{P}[1]_{\rm E}$	$I_2^{\rm D}/I_2^{\rm E}$	$I_1^{\rm D}/I_1^{\rm E}$	$b_{\rm sel,E}(\Delta^{\rm prj}=10)$	$b_{\rm sel,D}(\Delta^{\rm prj}=10)$
20	0.30	1.0000	1.0042	1.0041	45.9	45.8
20	0.50	1.0000	1.0040	1.0053	46.1	46.0
20	0.70	1.0001	1.0054	1.0046	48.1	47.9
50	0.30	0.9999	1.0054	1.0048	22.2	22.1
50	0.50	1.0000	1.0105	1.0059	20.8	20.6
50	0.70	0.9999	1.0028	1.0030	20.6	20.6

Headlines.

- At $N_z = 80$, Option D beats Option E on I_1, I_2: -0.47% and -0.32% vs -0.86% and -0.81%. No time penalty (both $\sim 22\,$ms).
- At $N_z = 40$, Option D is *worse* than Option E: -2.13% vs $+0.57\%$. D needs enough nodes to resolve the broad bell-shape it creates; with only ~ 20 GL nodes per side it under-samples the peak region.
- **Both schemes oscillate non-monotonically in N_z.** Neither is uniformly better across

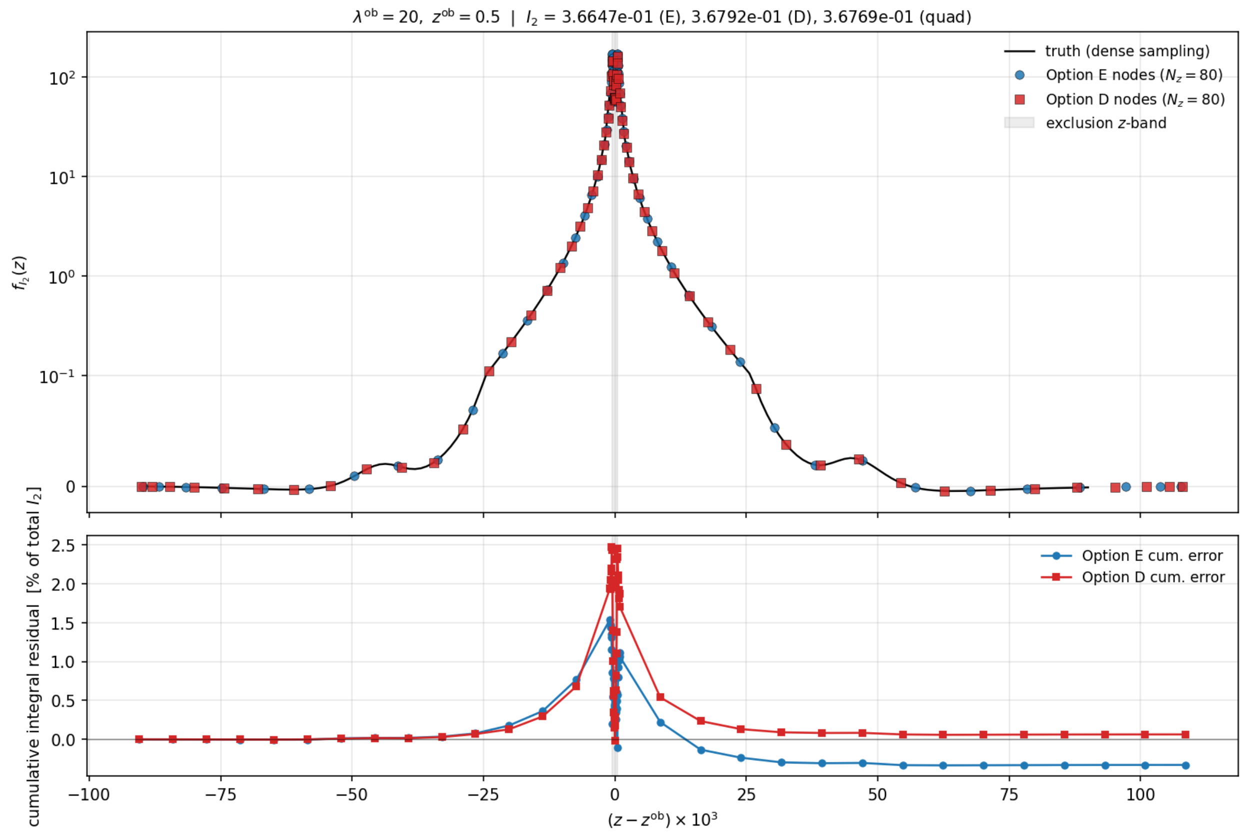

Figure 4: Where the nodes live and how each scheme's cumulative integral drifts. *Top:* $f_{I_2}(z)$ truth (black, densely sampled) with the Option E (blue circles) and Option D (red squares) nodes at $N_z = 80$ overlaid. Both schemes cover the double-peaked exclusion structure at $z = z^{\mathrm{ob}} \pm \delta z_{\mathrm{excl}}$ densely; E places its ring-plateau nodes on uniform z, D concentrates them via the compression, so the two schemes' node distributions differ visibly inside the exclusion band. *Bottom:* cumulative integral residual $\left[\sum_{z_i<z} w_i f(z_i) - \int_{z_{\mathrm{lo}}}^{z} f\,dz\right]/I_2^{\mathrm{quad}}$ (in percent; the reference $\int_{z_{\mathrm{lo}}}^{z} f\,dz$ is built piecewise from `scipy.integrate.quad` with rtol$=10^{-5}$). Both schemes accumulate a positive excess (up to $+2\%$) crossing the exclusion peaks, then relax toward their asymptotic bias (E: -0.33%, D: $+0.06\%$) as the outer tail integrates out.

all node counts. The residual ~ 0.3–1% is a floor of the fixed-rule methods; for sub-0.3% precision, use `scipy.integrate.quad` adaptively (~ 5 s per precompute).

- $\mathcal{P}[1]$ **is essentially identical between the two schemes**, differing only at the 10^{-4} level. Both handle the non-ξ_{NL} integrand fine.

Why not make Option D the default.

1. **Kernel-specific.** The transform is tuned to the $\xi_{\mathrm{NL}} \sim r^{-1.7}$ near-exclusion behaviour via the universal pivot R_{excl}. If ξ_{NL} is replaced by a linear ξ, a different halofit prescription, or an emulator, the location of the peak in t-coordinates shifts and node density is no longer optimal at $t = 0.881$. Option E makes no such assumption.
2. $\mathcal{P}[1]$ **has no** ξ_{NL}**,** so the transform's smoothing is wasted there. Option E's region split

correctly recognises that $\mathcal{P}[1]$ is smooth everywhere and uses GL directly.

3. **Non-monotonic convergence.** Both schemes oscillate, but D's oscillation amplitude is larger at low N_z ($\sim 2\%$ at $N_z = 40$) than E's ($\sim 0.8\%$). Option E is more forgiving of a user choosing a tight N_z for speed.

4. **Option E matches Matteo's notebook structure.** His `numerator2`/`int_over_phi4` follow the log-spaced LoS pattern; Option E is a drop-in upgrade.

When to use Option D. When MCMC runtime is dominated by I_1, I_2 and the user has already established $N_z \geq 60$ via convergence testing, Option D gives a $\sim 2\times$ reduction in residual error at the same node count. In `RichnessSelection` it is enabled via `SelBias(..., z_scheme='D')`; the default is `z_scheme='E'`.

4 Implementation notes

Both schemes are implemented in `src/richness_selection/sel_bias.py`, dispatched by the constructor argument `z_scheme` (`'E'` or `'D'`).

Option E, `_z_grid_option_E`, splits N_z as $N_{\rm ring} = \max(9,\, N_z/4)$ and $N_{\rm outer} = \max(15, (N_z - N_{\rm ring})/2)$ per side.

Option D, `_z_grid_option_D`, uses $N_{\rm side} = \max(15,\, N_z/2)$ GL nodes per foreground/background side in $t = \operatorname{asinh}(|\Delta\chi_\parallel|/R_{\rm excl})$, with no region split.

Both handle edge cases where $R_{\rm excl}$ exceeds the photo-z support or the ring band is clipped by $w_z = 0$. Two regression tests live in the `RichnessSelection` package:

- `validations/quad_validate.py`: compares one $N_z = 80$ result to `scipy.integrate.quad` at rtol 10^{-5}; must pass $< 1\%$ on $\mathcal{P}[1], I_1, I_2$.

- `validations/compare_schemes.py`: prints the table above (D vs E at $N_z \in \{40, 80, 160, 320\}$) at the reference point and across six $(\lambda^{\rm ob}, z^{\rm ob})$ bins.

For typical DES Y3 cluster redshift bins ($z^{\rm ob} \in [0.1, 0.8]$, $\lambda^{\rm ob} \in [20, 500]$) no retuning of $(N_{\rm ring}, N_{\rm outer})$ or $N_{\rm side}$ defaults should be needed: the *geometry* of Fig. 1 is universal in shape, only $R_{\rm excl}$ and $c/H(z^{\rm ob})$ shift with cosmology/richness, and both schemes track those scales explicitly.